SoccerGMR Performance Benchmark: Baseline vs. SI-Bind (E6) vs. SI-Bind + GEQ (Ours)

Head-to-head evaluation across 9 temporal DETR backbones on Standard Test (1,036 queries) and Full-Decodable Test (2,970 queries). Strict single seed (2023).

		Standard Test Split (1,036 queries)				Full-Decodable Test Split (2,970 queries)			
Backbone	**Configuration**	**mAP (%)**	**Orc-Gap (%)**	**Rej-F1@.5 (%)**	**AUROC (%)**	**mAP (%)**	**Orc-Gap (%)**	**Rej-F1@.5 (%)**	**AUROC (%)**
Moment-DETR	1. GMR Baseline	6.68	24.28	67.94	70.20	5.47	23.48	72.08	78.91
	2. GMR + SI-Bind (E6)	**15.21**	23.66	65.52	**74.32**	12.58	22.21	68.83	**81.86**
	3. GMR + SI-Bind (E6) + GEQ (Ours)	14.97	**16.61**	**75.33**	74.23	**12.95**	**15.95**	**78.51**	81.81
EaTR	1. GMR Baseline	16.49	21.51	65.60	76.56	15.19	20.39	67.38	81.59
	2. GMR + SI-Bind (E6)	**17.12**	21.80	67.74	76.23	**15.35**	20.89	68.66	81.03
	3. GMR + SI-Bind (E6) + GEQ (Ours)	15.54	**17.51**	**72.25**	**76.57**	14.36	**16.42**	**76.22**	**83.05**
QD-DETR	1. GMR Baseline	14.62	20.23	71.48	**77.01**	12.65	19.05	72.49	81.75
	2. GMR + SI-Bind (E6)	16.30	20.65	73.71	76.63	13.20	19.84	75.91	82.47
	3. GMR + SI-Bind (E6) + GEQ (Ours)	**18.27**	**15.94**	**75.14**	76.16	**14.98**	**15.70**	**78.45**	**82.73**
CG-DETR	1. GMR Baseline	**18.65**	25.66	57.47	**76.40**	14.67	23.28	63.23	81.81
	2. GMR + SI-Bind (E6)	17.38	22.89	70.85	75.92	**15.54**	21.61	73.14	80.94
	3. GMR + SI-Bind (E6) + GEQ (Ours)	17.66	**16.30**	**75.64**	**76.40**	15.47	**15.74**	**78.88**	**82.36**
BM-DETR	1. GMR Baseline	4.00	17.59	**75.00**	**71.55**	3.65	17.62	78.58	79.11
	2. GMR + SI-Bind (E6)	**7.72**	20.91	73.40	70.18	**7.03**	20.30	77.47	**79.27**
	3. GMR + SI-Bind (E6) + GEQ (Ours)	6.17	**13.19**	**75.00**	70.57	5.63	**13.44**	**78.86**	79.08
BAM-DETR	1. GMR Baseline	23.25	19.82	70.43	**78.66**	20.62	18.37	71.79	80.99
	2. GMR + SI-Bind (E6)	23.78	20.40	68.96	78.37	**21.78**	19.07	69.28	78.69
	3. GMR + SI-Bind (E6) + GEQ (Ours)	**23.99**	**17.48**	**71.31**	77.08	21.56	**16.37**	**73.81**	**83.35**
Sim-DETR	1. GMR Baseline	20.73	22.51	74.69	72.97	19.18	21.78	**75.72**	78.87
	2. GMR + SI-Bind (E6)	**21.17**	21.26	74.89	75.12	**20.08**	20.83	73.10	77.83
	3. GMR + SI-Bind (E6) + GEQ (Ours)	21.14	**12.72**	**75.00**	**75.60**	19.96	**12.59**	75.25	**81.21**
UVCOM	1. GMR Baseline	7.12	17.24	75.00	70.03	6.40	16.83	**76.80**	78.98
	2. GMR + SI-Bind (E6)	**18.99**	**16.30**	**75.40**	76.91	**17.74**	**15.55**	73.11	75.20
	3. GMR + SI-Bind (E6) + GEQ (Ours)	17.94	17.89	72.92	**77.37**	16.55	17.04	73.98	**81.37**
FlashVTG	1. GMR Baseline	**25.43**	**25.47**	**66.35**	**74.30**	**20.43**	**24.13**	**68.48**	**82.21**
	2. GMR + SI-Bind (E6)		*-- running / pending --*				*-- running / pending --*		
	3. GMR + SI-Bind (E6) + GEQ (Ours)		*-- running / pending --*				*-- running / pending --*		

Method 1 (GMR Baseline): Official temporal DETR baseline checkpoints evaluated with native bipartite matching.
Method 2 (GMR + SI-Bind [E6]): Full Structural Triad (Temporal Residual Stream + Slot-Private Innovation + Dominance-Aware Refresh).
Method 3 (GMR + SI-Bind [E6] + GEQ [Ours]): Two-stage Global Event Query (GEQ) fine-tuning selected by Validation Orc-gap with 20 epochs patience.
Orc-Gap (%): Area between Oracle envelope and G-mIoU@1 threshold curve across tau in [0, 1] (Integral of Oracle_G1 minus G1-AUC). Lower is better.
Bold values highlight the optimal performance across all 3 configurations per backbone. FlashVTG E6 currently in queue.